

Shenzhen Elida Technology Co.,Limited

APPROVAL SHEET

承 认 书

客户名称 Customer	
产品型号 Part NO.	KD35T133-A
产品内容 Product type	Mode: Transmissive type .Normally Black TFT LCD Module LCD Module: Graphic 320RGB*480Dot-matrix
备注栏 Remarks	☐ APPROVAL FOR SEPCIFICATIONS ONLY ☒ APPROVAL FOR SEPCIFICATIONS AND SAMPLE
客户确认签章 Signature by Customer:	
备注 Notes:	

PREPARED BY	CHECKED BY	APPROVED BY

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1. General Description

KD35T133-A is a 320*480 dots matrix TFT LCD module. It has a TFT panel composed of 960sources and 480gates. The LCM can be easily accessed by micro-controller.

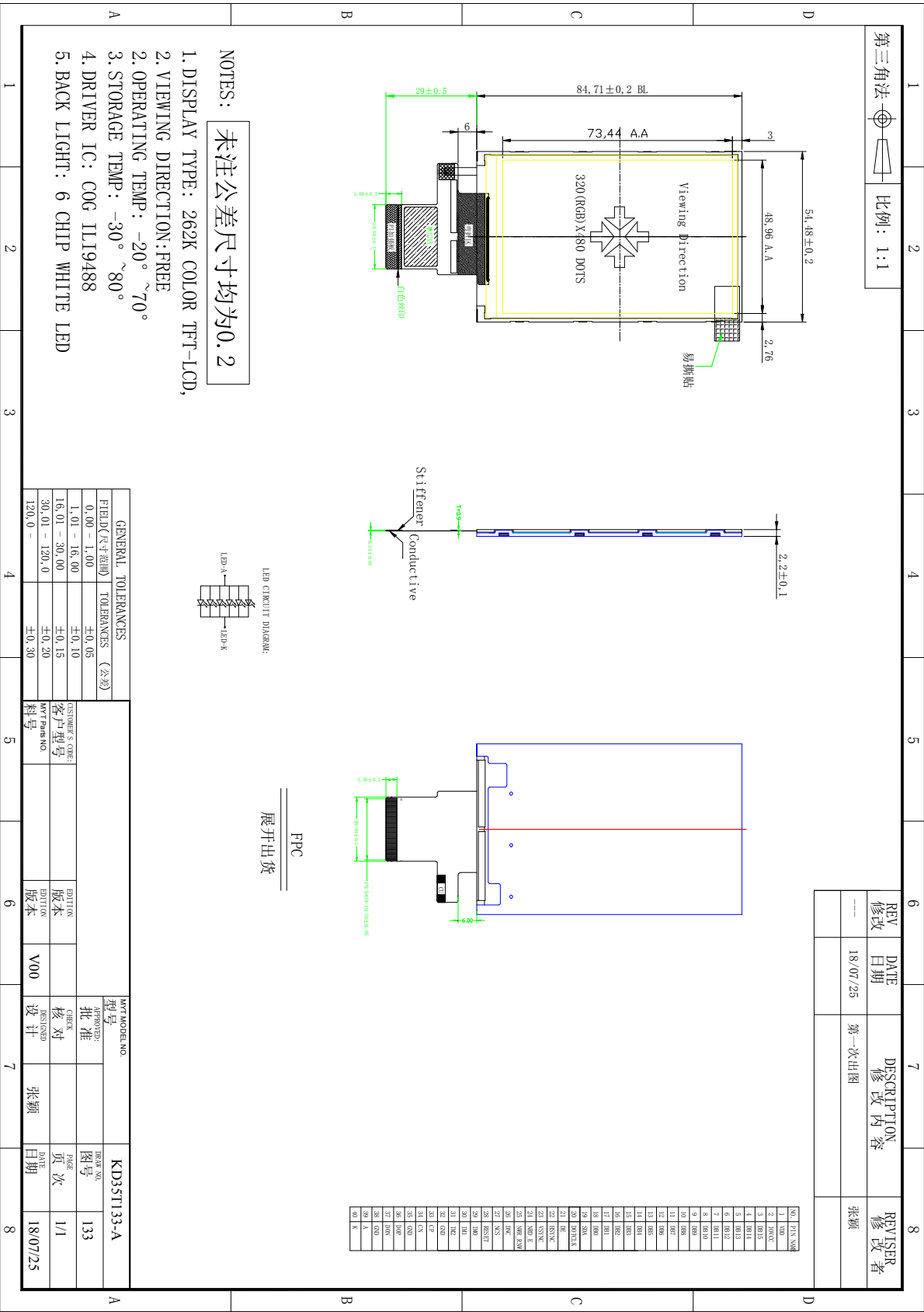
2. Features

Display Mode	Transmissive
	a-TFT
Display Format	Graphic 320RGB*480Dot-matrix
Input Data	RGB/MCU/MIPI
Viewing Direction	ALL
Drive	ILI9488

3. Mechanical Specification

Item	Specifications	Unit
Dimensional outline	54.48(H) x 84.71 (V) x2.20(T) (FPC not include)	mm
Resolution	320RGB*480	dots
LCD Active area	48.96(H) x 73.44(V)	mm
TP Active area	-	mm
Pixel size	0.153(H) x 0.153(V)	mm

4.MechanicalDimens



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5. Maximum Ratings

Item	Symbol	Min	Max	Unit	Remark
Supply voltage	IOVCC	1.65	3.3	V	
Analog Supply Voltage	VCI	2.5	3.3	V	
Operating Temperature	T _{ORT}	-20	70	℃	
Storage temperature	T _{STG}	-30	80	℃	

6. Electrical Characteristics

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Supply voltage	Logic	V _{CC}		2.7	2.8	2.9	V
Input Voltage	H level	T _{IH}		0.8*IOVCC		IOVCC	V
	L level	T _{IL}		-0.3		0.2* IOVCC	
Storage temperature		I _{DD}	With internal voltage generation V _{CC} =2.8V; T _{emp} =25℃			TBD	mA

7. Backlight Characteristic

Item	Symbol	Min	Typical	Max	Unit
LED module Forward voltage	V _{LED}		3.2		V
LED module current	V _{LED}		120		mA
L/G Surface Luminance ★1	L _S		8000		Cd/m ³
LCM Surface brightness uniform ★2	L _D	80			%

★ 1Test condition is:

(a) Center point on active area.

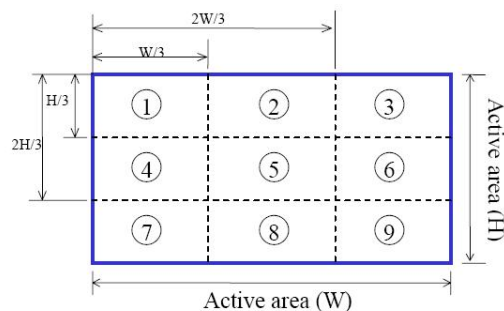
(b)Best Contrast.

★2Uniform measure condition:

(1)Measure 9 point. Measure location show below;

(2)Uniform=(Min. brightness /Max. brightness)*100%

(3)Best Contrast.



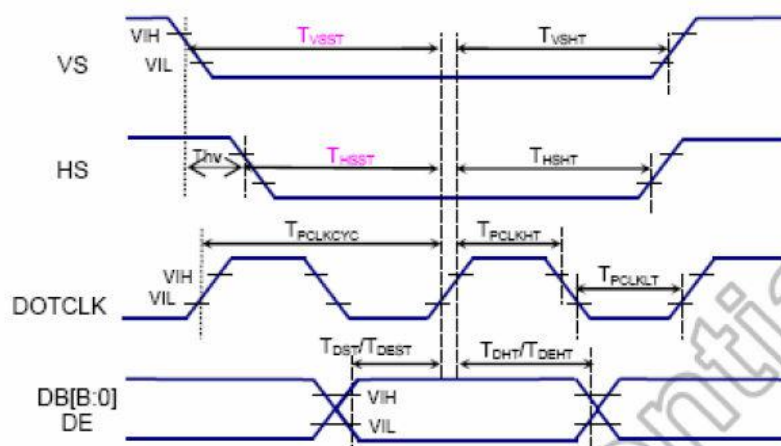
8. Module Function Description

8.1 Pin Descriptions

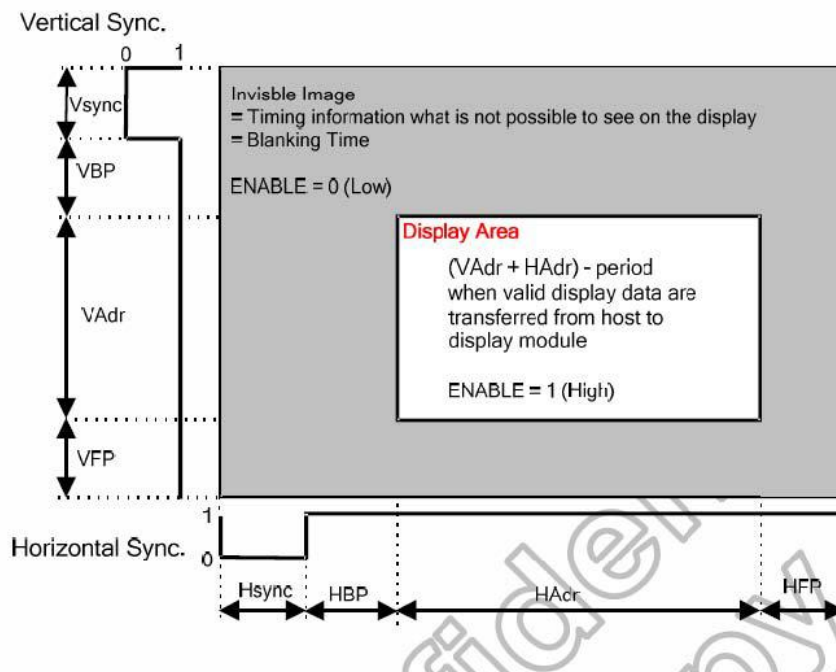
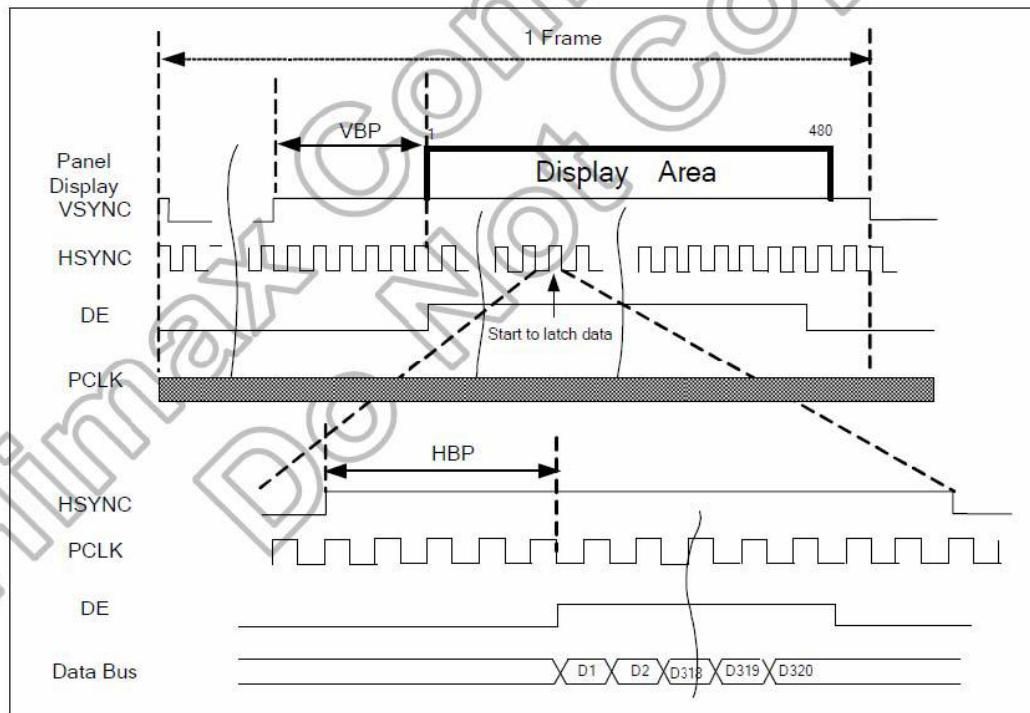
Pin No	Symbol	Description
1	VDD	Power Supply For LCD 2.8V
2	IOVCC	Power Supply For LCD 1.8V/2.8V
3-18	DB15-DB0	DATA BUS
19	SDA	serial data input/output bi-direction pin
20	DOTCLK	Dot clock signal
21	DE	Data enable signal
22	HSYNC	Horizontal synchronizing signal in DPI interface. Let to open or connected to VSSD.
23	VSNC	Vertical synchronizing signal in DPI interface. Let to open or connected to VSSD
24	NRD_E	serve as a read signal
25	NWR	WRX pin, serves as a write signal,SCL pin as Serial Clock when operates in the serial interface
26	NDC	Data/Command Selection pin
27	NCS	Chip Select PIN
28	RESET	Reset Signal
29	IMO	Interface Mode Select.(See IC DATASHEET)
30	IM1	Interface Mode Select.(See IC DATASHEET)
31	IM2	Interface Mode Select.(See IC DATASHEET)
32	GND	System Ground
33	CP	Positive polarity of low voltage differential clock signal
34	CN	Negative polarity of low voltage differential data signal
35	GND	System Ground
36	D0P	Positive polarity of low voltage differential data signal
37	D0N	Negative polarity of low voltage differential data signal
38	GND	System Ground
39	A	Power Supply For LED Backlight Anode Input
40	K	Power Supply For LED Backlight Cathode Input

8.2 Timing characteristics.

DPI interface characteristics



Item	Symbol	Condition	Spec.			Unit
			Min.	Typ.	Max.	
Pixel low pulse width	T_{CLKLT}		15	-	-	ns
Pixel high pulse width	T_{CLKHT}		15	-	-	ns
Vertical Sync. set-up time	T_{VSST}		15	-	-	ns
Vertical Sync. hold time	T_{VSHT}		15	-	-	ns
Horizontal Sync. set-up time	T_{HSST}		15	-	-	ns
Horizontal Sync. hold time	T_{HSHT}		15	-	-	ns
Data Enable set-up time	T_{DEST}		15	-	-	ns
Data Enable hold time	T_{DEHT}		15	-	-	ns
Data set-up time	T_{DST}		15	-	-	ns
Data hold time	T_{DHT}		15	-	-	ns
Phase difference of sync signal falling edge	T_{hv}		0	-	320	Dotclk



Power on/off sequence

Power source IOVCC, VCI can be applied and powered down in any order.

IOVCC, VCI can be powered down in any order.

During power off, if LCD is in the Sleep Out mode, IOVCC, VCI must be powered down minimum 120msec after NRESET has been released.

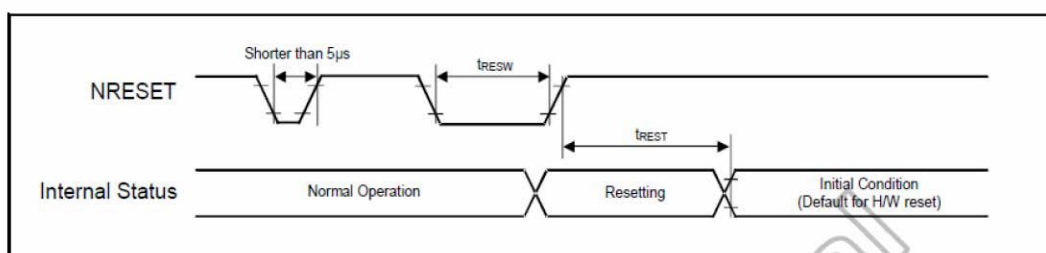
During power off, if LCD is in the Sleep In mode, IOVCC, VCI can be powered down minimum 0msec after NRESET has been released.

NCS can be applied at any timing or can be permanently grounded. NRESET has priority over NCS.

- Note:** (1) There will be no damage to the display module if the power sequences are not met.
 (2) There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.
 (3) There will be no abnormal visible effects on the display between end of Power on Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.
 (4) If NRESET line is not held stable by host during Power on Sequence as defined in Sections 5.11.1.1 and 5.11.1.2, then it will be necessary to apply a Hardware Reset (NRESET) after Host Power on Sequence to ensure correct operation. Otherwise correct function is not guaranteed.

If NRESET line is not held stable by host during Power on Sequence as defined in Sections 5.9.1.1 and 5.9.1.2 then it will be necessary to apply a Hardware Reset (NRESET) after Host Power on Sequence is complete to ensure correct operation, otherwise correct functionality is not guaranteed. The power on/off sequence is illustrated as below

Reset Timing Characteristics



Symbol	Parameter	Related Pins	Spec.			Note	Unit
			Min.	Typ.	Max.		
tRESW	Reset low pulse width ⁽¹⁾	NRESET	10	-	-	-	µs
tREST	Reset complete time ⁽²⁾	-	5	-	-	When reset applied during SLPIN mode	ms
		-	120	-	-	When reset applied during SLPOUT mode	ms

9. Electro-optical Characteristics

9.1 Optical specification

Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Note
Transmittance (without Polarizer)		T (%)	—	—	(14.74)	—	—	
Contrast Ratio		CR	$\theta=0$	400	500	—	—	(1)(2)
Response time	Rising	T _R	Normal viewing angle	—	(4)	(8)	msec	(1)(3)
	Falling	T _F		—	(12)	(24)		
				—				
Color gamut		S(%)			(60)		%	
Color chromaticity (CIE1931)	White	W _x		(0.283)	(0.303)	(0.323)		(1)(4) CF glass
		W _y		(0.305)	(0.325)	(0.345)		
	Red	R _x		(0.606)	(0.626)	(0.646)		
		R _y		(0.314)	(0.334)	(0.354)		
	Green	G _x		(0.257)	(0.277)	(0.297)		
		G _y		(0.529)	(0.549)	(0.569)		
	Blue	B _x		(0.122)	(0.142)	(0.162)		
		B _y		(0.102)	(0.122)	(0.142)		
Viewing angle	Hor.	θ_L	CR>10	70	80	—		Viewing Angle base on using EWV Polarizer Reference Only
		θ_R		70	80	—		
	Ver.	θ_U		70	80	—		
		θ_D		70	80	—		
Optima View Direction		ALL						(5)

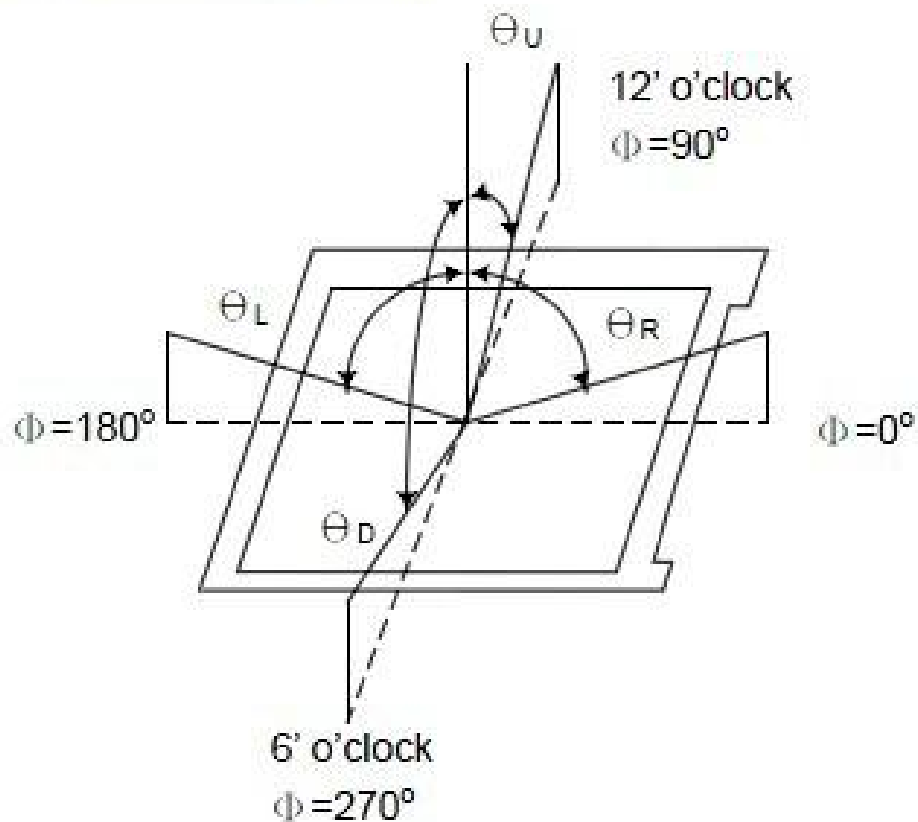
9.2 Measuring Condition

- _ Measuring surrounding : dark room
- _ Ambient temperature : $25 \pm 2^\circ\text{C}$
- _ 15min. warm-up time.

9.3 Measuring Equipment

- _ FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-5A for other optical characteristics.

Note (1) Definition of Viewing Angle :

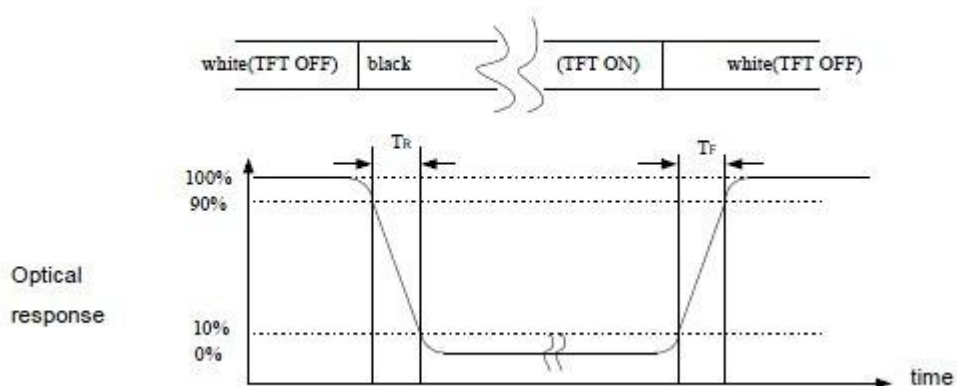


Note (2) Definition of Contrast Ratio(CR) :
measured at the center point of panel

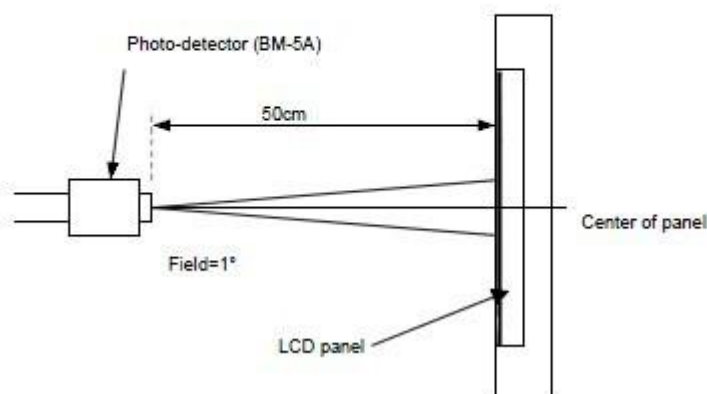
$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

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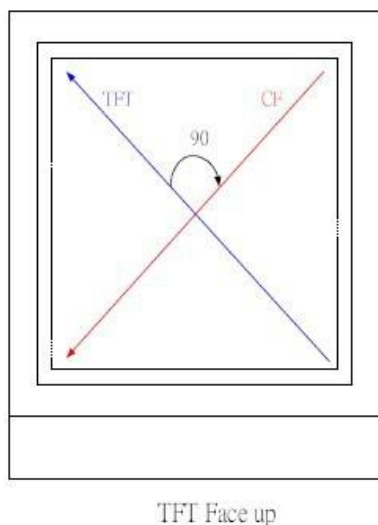
Note (3) Definition of Response Time : Sum of T_R and T_F



Note (4) Definition of optical measurement setup



Note (5) Rubbing Direction (The different Rubbing Direction will cause the different optima view direction).



10. Reliability

10.1Mtbf

The LCD module shall be designed to meet a minimum MTBF value of 50000 hours with normal

10.2Test condition

NO.	ITEM	CONDITION	CRITERION
1	High Temperature Non-Operating Test	80℃* 96 Hrs	。 No Defect Of Operational Function In Room Temperature Are Allowable
2	Low Temperature Non-Operating Test	-30℃* 96 Hrs	
3	High Temperature/Humidity Non Operating Test	60℃*90%RH* 96Hrs	
4	High Temperature Operating Test	70℃* 96Hrs	。 IDD of LCM in Pre-and Post-Test Should Follow Specification
5	Low Temperature Operating Test	-20℃* 96 Hrs	
6	Thermal Shock Test	-20℃ (30Min) v 70℃ (30Min) *10CYCLES	

Notes:

1. Judgments should be made after exposure in room temperature for two hours.
2. The distill water is used for the high temperature/humidity test.
3. The sample above is individually for every reliability tests condition.

11.Inspection standards

1.AQL(Acceptable Quality Level

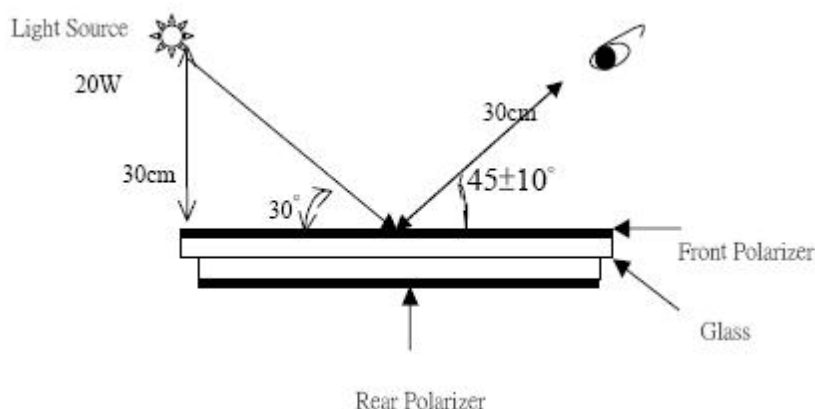
AQL of major and minor defect.

	MAJOR DEFECT	MINOR DEFECT
AQL	0.65	1.5

2. Basic conditions for inspection

The LCM face to us, in normal environment, the lux is 1000 ± 200 . (Darkroom's lux: 100 ± 50), About an angle of incidence 30°, a distance of 30 cm with an angle of 45 degree to check the products without uncovering the film!

(As shown below)

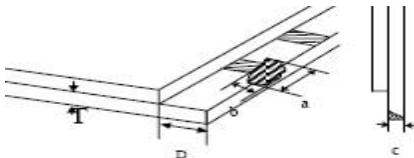
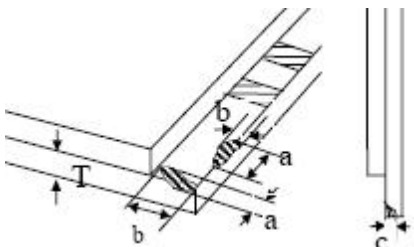
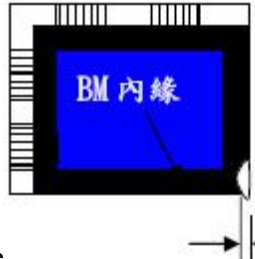


3.Inspection item and criteria

3.1 Visual inspection criterion in immobility

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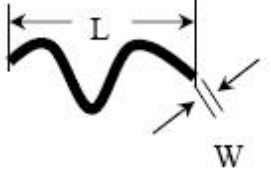
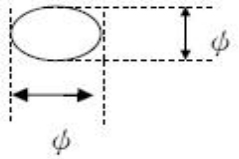
3.1.1Glass defect

NO	Defect item	Criteria	Remark
1	Dimension Unconformity (Major defect)	By Engineering Drawing	
2	Cracks (Major defect)	1. Linear cracks panel 2. Nonlinear crack contrast by limited sample	
3	Glass extrude the conductive area (minor defect)	a: disregards and no influence assemblage. 1) $b \leq 1/3$ Pin width(non bonding area) 2) bonding area ≤ 0.5 mm	A: Length, b: Width
4	Pin-side ,conductive area damaged (minor defect)	(a c: disregards) $b \leq 1/3$ of effective length for bonding electrode	a: length, b: Width, c: Thickness 
5	Pin-side,non-conductive area damaged (minor defect)	1) Damage area don't touch the ITO (Inclueing contraposition mark, except scribing mark) 2) $C < T$ $b \leq BM 1/3$ of width 3) $c = T$ b not touch the seal glue 4) a disregards	a: Length, b: Width c: Thickness 
6	Non-pin-side damage (minor defect)	$c < T$ 1) b exceeds $1/3 BM$ $c = T$ b not touch the seal glue	c: Thickness b: width of 

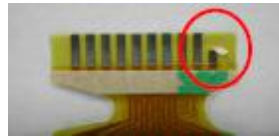
3.1.2LCD appearance defect(View area)

NO	Defect item	Criteria		Remark
1	Fiber、 glass cratch、 polarizer	Specification	Allowable	note1:L: Length, W: Width
		$W \leq 0.03$ mm	disregard	note2: disregard if out of AA

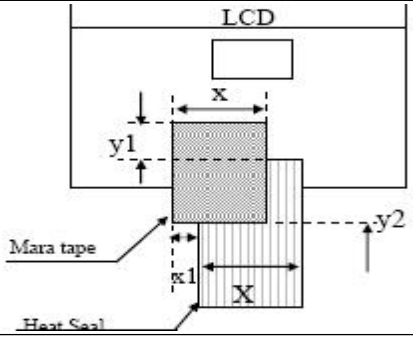
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	scratch/folded (minor defect)	$0.03\text{mm} < W \leq 0.05\text{mm};$ $L \leq 3.0\text{mm}$	2	
		$0.05\text{mm} < W \leq 0.1\text{mm};$ $L \leq 3.0\text{mm}$	1	
		$W > 0.1\text{mm}; L > 3.0\text{mm}$	0	
2	Polarizer bubble、 concave and convex (minor defect)	$\phi \leq 0.2\text{mm}$	disregard	note1: $\phi = (L+W)/2$, L:Length, W :Width note2:disregard if out of AA
		$0.2\text{mm} < \phi \leq 0.3\text{mm}$	2	
		$0.3\text{mm} < \phi \leq 0.5\text{mm}$	1	
		$0.5\text{mm} < \phi$	0	
3	Black dots、dirty dots、 impurities、eye winker (minor defect)	$\phi \leq 0.15\text{mm}$	disregard	note2:disregard if out of AA 
		$0.15\text{mm} < \phi \leq 0.25\text{mm}$	2	
		$0.25\text{mm} < \phi \leq 0.3\text{mm}$	1	
		$0.3\text{mm} < \phi$	0	
4	Polarizer prick (minor defect)	$\phi \leq 0.1\text{mm}$	disregard	note1: $\phi = (L+W)/2$, L=Length, W=Width note2:the distance between two dots>5mm
		$0.1\text{mm} < \phi \leq 0.25\text{mm}$	3	
		$\phi > 0.25\text{mm}$	0	

3.1.3FPC

NO	Defect item	Criteria		Remark
1	Copper screen peel (minor defect)	Copper screen peel 【Reject】		
2	No release tape or peel	No release tape or peel 【Reject】		
3	Dirty dot and impurity of FPC for customer using side (minor defect)	Specification	Allowable	Note1: Cannot have stride ITO impurities
		$\phi \leq 0.25\text{mm}$	2	
		$\phi > 0.25$	0	

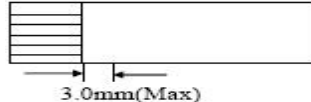
3.1.4Black tape & Mara tape

NO	Defect item	Criteria	Remark
1	FPC or H/S black tape (minor defect)	1. shift spec: 1) glue to the polarize 【Reject】 2) IC bare 【Reject】 2. left-and-right spec: 1) exceed of FPC edge or H-S edge 【Reject】 2) IC bare 【Reject】	
2	No black tape	No black tape	

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	(major defect)	【Reject】	
3	Tape position mistake (minor defect)	Not by engineering drawing	
4	Mara tape defect (minor defect)	Peel before pulling the protecting film 【Reject】	

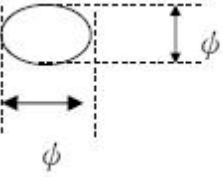
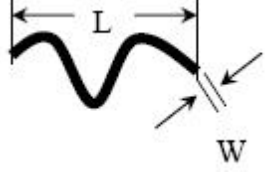
3.1.5 Silicon and Taffy glue

NO	Defect item	Criteria	Remark
1	Quantity of silicon (major defect)	Uncover the ITO and circuit area 【Reject】	note: compared by engineering
2	Taffy glue (major defect)	1.Uncover the reveal copper area【Reject】 2.Cover layer 0.3mm(Min)~3.0mm(Max) 【Reject】	note: if customer has special requirement, refer to the technical document 
3	Depth of glue covering (major defect)	Depth of glue covering overtop front Polarizer 【Reject】	Except of the special requirement

3.2Electrical criteria

NO	Defect item	Criteria	Remark
1	No display (major defect)	No display 【Reject】	
2	Missing line (major defect)	Missing line 【Reject】	
3	Seg-com light and dark (major defect)	Seg-com light and dark 【Reject】	ND filter 2% test
4	No display in immobility (major defect)	No display in immobility 【Reject】	
5	Flicker of Pattern (major defect)	Flicker of Pattern 【Reject】	
6	Mura (major defect)	ND filter 2%test	
7	Over current (major defect)	Over current 【Reject】	
8	Voltage out of specification (major defect)	Voltage out of specification 【Reject】	
9	Pattern blur, error code (major defect)	Pattern blur, error code 【Reject】	
10	Dark light, Flicker	Dark light, Flicker	

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	(major defect)	【Reject】		
11	Black/white dots 、 Dirty dots、 eye winker (major defect)	Specification	Allowable	Note1:disregard if out of AA 
		$\phi \leq 0.15\text{mm}$	disregard	
		$0.15\text{mm} < \phi \leq 0.25\text{mm}$	2	
		$0.25\text{mm} < \phi \leq 0.3\text{mm}$	1	
		$0.3\text{mm} < \phi$	0	
12	Fiber、glass crutch、Polarizer scratch/folded (major defect)	$W \leq 0.03\text{mm}$	disregard	Note1:L: Length, W: Width Note2: disregard if out of AA 
		$0.03\text{mm} < W \leq 0.05\text{mm}$ $L \leq 3.0\text{mm}$	2	
		$0.05\text{mm} < W \leq 0.1\text{mm}$ $L \leq 3.0\text{mm}$	1	
		$W > 0.1\text{mm}; L > 3.0\text{mm}$	0	

12.Precautions for using LCD modules.

12.1 Safety

- (1)Do not swallow any liquid crystal ,even if there is no proof that liquid crystal is poisonous.
- (2)If the LCD panel breaks, be careful not to get liquid crystal to touch your skin.
- (3)If skin is exposed to liquid crystal, wash the area thoroughly with alcohol or soap.

12.2Storage Conditions

- (4)Store the panel or module in a dark place where the temperature is $23 \pm 5^{\circ}\text{C}$ and the humidity is below $45 \pm 20\%\text{RH}$.
- (5)Store in anti-static electricity container.
- (6)Store in clean environment, free from dust, active gas, and solvent.
- (7)Do not place the module near organics solvents or corrosive gases.
- (8)Do not crush, shake, or jolt the module.

12.3Handling Precautions

- (9)Avoid static electricity, which can damage the CMOS LSI.
- (10)The polarizing plate of the display is very fragile, please handle it very carefully.
- (11)Do not give external shock.
- (12)Do not apply excessive force on the surface.
- (13)Do not wipe the polarizing plate with a dry cloth, as it may easily scratch the surface of plate.
- (14)Do not use ketonic solvent & Aromatic solvent, use with a soft cloth soaked with a cleaning naphtha solvent.
- (15)Do not operate it above the absolute maximum rating.

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(16)Do not remove the panel or frame from the module.

12.4Warranty

The period is within twelve months since the date of shipping out under normal using and storage conditions.

13.Factory

FACTORY NAME:

FACTORY ADDRESS:

FACTORY PHONE:

14.Revision history

Version	Revise record	Date
A01	Original version	2016-6-21